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Weiqi Wang

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SUMMARY

- Data scientist (Ph.D. Applied Math) with end-to-end ownership: frame metrics, build interpretable models, validate rigorously, and ship reproducible Python code.
- Hands-on with Python (pandas, NumPy, scikit-learn, matplotlib), SQL, Git/Linux/Jupyter/pytest. Experience on 1M-row forecasting, experimentation/benchmarking, and time-series ETL.

SKILLS

Programming	Python (pandas, NumPy, scikit-learn, matplotlib)
Data/DB	SQL, CSV/Parquet pipelines
Tools	Git, Linux, Jupyter, pytest
ML/Stats	Time-series forecasting, probabilistic/Markov models, Neural networks

EDUCATION

Ph.D. in Mathematics, University at Buffalo 2016.08 – 2021.08

Thesis: Numerical Solution to the Corner Angle Problem for an Elastic Solid with Anisotropic Surface Energy

B.S. in Mathematics, Fudan University 2011.09 – 2016.06

PROJECTS

Railway Delay Prediction 2022.09 – 2023.05

- Built an **interpretable non-homogeneous Markov** forecaster; verified first-order dependence via **chi-square** test; recovered sparse transitions with **Gaussian-kernel** smoothing.
- Defined decision metrics and delivered reproducible Python pipelines on **1M rows**.

Baseball Game Outcome Modeling (team project) 2023.09 – 2023.10

- Cleaned/joined Retrosheet + 538 Elo; built season-aware simulations and **logistic models**. Delivered a lower Brier score than 538 Elo (≈ 0.2404 vs ≈ 0.2436); a simple opponent baseline was ≈ 0.2503 .

AoI Grid Patrol (Optimization & Simulation) 2023.12 – 2024.09

- Designed and benchmarked routing algorithms to minimize **Age of Information**; built Python evaluation harness over **1k+ randomized grids** with controlled seeds.
- Reported mean/percentile AoI and sensitivity to grid size/robot speed; produced plots/tables used in publication.

Side-Channel Analysis (DPA v4 Contest) 2014.08 – 2016.06

- Applied **PCA denoising** and trace alignment; implemented **template attacks** (Python/MATLAB); small model changes improved key-recovery accuracy.
- Collaborated with a security lab; results recognized on the contest *Hall of Fame*.

EXPERIENCE

Postdoctoral Fellow, University of Victoria 2023.11 – present

- Reproducible ML experiments (JAX): orchestrated training/evaluation pipelines, benchmarking against numerical baselines; documentation and unit tests.

NumerixS Quant 2023.06 – 2023.07

- Reduced false alerts in market-data integrity checks (kdb+/q); built a small replay harness and wrote a short runbook for users.

Postdoctoral Fellow, Concordia University 2021.09 – 2023.05

- Designed algorithms and Python evaluation harnesses for graph/network problems (Age of Information); emphasis on metrics, experiments, and reporting.

SELECTED PUBLICATIONS (DATA-SCIENCE RELEVANT)

- *Approximation schemes for AoI minimization in UAV grid patrols*. Ad Hoc Networks, 2025.
- *A novel Markov model for near-term railway delay prediction*. Computers & Industrial Engineering, 2023.